

SYSTEM INTEGRATION AND ARCHITECTURE

MONTHLY PROGRESS REPORT

Student Information

DATE SUBMITTED

June 17, 2026

MPR NO:

4

SECTION

BSINFOTECH 3-3C

PROJECT TITLE

E-Serbisyo: Web-Based Document Request and Information System using QR Code Authentication

STUDENTS NAME

1. CHUA, REYMART M. - Project Leader
2. KAMILAN, DIANA F. - Documentation Leader
3. MORALES, JOHN REY F. - Technical Leader
4. TANA, LARA MHAY CLAIRE L. - Communication Leader
5. TOMAS, TJ C. - Quality Assurance Leader
6. REGUALOS, CHRIS TOKIYO T. - Communication Leader

1. Tasks Planned for the Month

- Complete and finalize all remaining features including Email Notification Service, QR Code Document Verification, and Automated Database Backup System
- Implement Forgot Password and Reset Password functionality for all user roles
- Develop System Settings and Super Admin Dashboard modules
- Create test scripts and comprehensive testing tools for QR verification and email services
- Prepare Cloudflare Tunnel configuration and documentation for production deployment readiness
- Finalize all system documentation including integration guide, testing manual, and deployment instructions
- Conduct end-to-end testing of all workflows across Resident, Secretary, Treasurer, Admin, and Super Admin roles

2. Tasks Accomplished

- Successfully implemented and tested Email Notification Service via Nodemailer SMTP (Gmail App Password), with automated email alerts for registration approval, payment confirmation, and document status updates
- Fully integrated QR Code Document Verification system allowing public verification via webcam scanning, file upload, and manual hash entry
- Completed Forgot Password and Reset Password functionality with secure token-based authentication and 1-hour expiry
- Developed and deployed Automated Database Backup system with ZIP archiving (SQL dump + encrypted uploads), manual creation/download/restore, and daily midnight cron job with 7-backup rotation
- Implemented System Settings module for Captain-only configuration management
- Built Super Admin Dashboard with financial overview, collection figures, and request metrics
- Created comprehensive test files: test-smtp.js for email testing, test-qr-seed.js for QR verification,

comprehensive-api-test.html for full API endpoint testing

- Configured and documented Cloudflare Tunnel setup for production deployment with custom domain routing (api.brgy143.gov.ph, portal.brgy143.gov.ph)
- Documented the complete system in SYSTEM_INTEGRATION_DOCUMENT.md, TESTING_MANUAL.md, and README_CLOUDFLARE_TUNNEL.md
- Finalized all Resident portal features: Dashboard, Request Wizard (4-step), My Requests with filtering, and Profile management
- Completed all Admin/Official modules: Requests Queue, Payments Queue, Manage Residents, Manage Documents, Manage Officials, Manage Announcements, Audit Logs, Backup Management, My Profile, and Force Password Change
- Achieved approximately 95-100% overall system development completion with all core features operational and tested

3. Current System Status

- The system is fully complete with 100% of planned features implemented and operational
- All major modules are integrated and functioning: Public Portal (Home, Registration, Login, QR Verification, Password Reset), Resident Portal (Dashboard, Request Wizard, My Requests, Profile), and Admin Portal (all role-based dashboards and management modules)
- Email notification system is fully operational with SMTP integration, sending automated alerts for all key workflow events
- QR Code verification system is working with three verification methods: webcam scanning, file upload, and manual hash entry
- Automated backup system is active with daily midnight cron job, manual backup creation, download, restore, and 7-backup rotation
- AES-256-CBC encryption is implemented for all file uploads, database fields, and digital signatures
- PDF generation with QR stamping and digital signature embedding is complete
- Cloudflare Tunnel configuration is documented and ready for production deployment
- All role-based access controls (Captain, Admin, Secretary, Treasurer) are properly enforced with JWT authentication
- The system is now in production-ready state, pending final deployment and User Acceptance Testing

4. Problems Encountered

- Email service requires Gmail App Password configuration and stable SMTP connection, which is dependent on the deployment environment
- QR code scanning via webcam may have limited functionality on older mobile browsers; fallback methods (file upload, manual entry) have been implemented as mitigation
- Domain acquisition for official barangay web presence is still in progress, which affects production deployment timelines
- Dynamic DNS and Technitum (TNO) service coordination requires further discussion with the barangay beneficiary for internet connectivity requirements
- Server hosting cost assessment and budget allocation have not yet been finalized with the project beneficiary
- Some UI/UX elements may require minor adjustments based on end-user feedback during UAT

5. Actions Taken / Solutions

- Prepared detailed deployment documentation including Cloudflare Tunnel setup, environment configuration, and server requirements
- Implemented multiple QR verification fallback methods (webcam, file upload, manual hash) to ensure broad device

compatibility

- Coordinated with the project beneficiary regarding domain name preferences and budget allocation for hosting services
- Researched and documented deployment cost estimates including domain registration (DNS), cloud hosting (VPS), and email service (SMTP)
- Prepared discussion points for Technitum (TNO) and Dynamic DNS service coordination with the barangay technical team
- Conducted thorough internal testing of all system features to ensure stability before deployment and UAT
- Updated and finalized all system documentation to reflect the current state of the system

6. Plan for Next Month

- Deploy the system to production environment using Cloudflare Tunnel with a custom domain for both the API backend and frontend portal
- Conduct real-time live testing of all system features in a production-like environment with actual users
- Perform comprehensive final testing of all features including: Resident registration and document request workflow, QR code verification end-to-end, Email notification delivery and formatting, Payment processing and PDF generation, Backup creation and restore functionality
- Coordinate domain acquisition (e.g., brgy143.gov.ph or similar) through the appropriate government domain registrar (DOST/GOV.PH)
- Discuss and finalize deployment service costs including: Domain registration and renewal fees, Cloud VPS or shared hosting monthly subscription, SMTP/email service plan (if applicable), SSL certificate and DNS management costs
- Coordinate server hosting talks with potential providers (e.g., Cloudflare, AWS, Google Cloud, local Philippine hosting providers like SkyCable, PLDT, or DOST ASTI)
- Evaluate and plan Dynamic DNS configuration for Technitum (TNO) integration to ensure stable internet connectivity for the barangay office
- Finalize User Acceptance Testing (UAT) with barangay officials and residents, collecting feedback for final refinements
- Prepare final project presentation materials and complete all remaining documentation requirements for project submission

Prepared by:

Diana F. Kamilan
Documentation Leader

Date: June 17, 2026

Checked By:

PROF. ERNANIE M. CARLOS JR., MIT
Subject Adviser

Date: June 17, 2026